

The *n*-queens Problem

Consider the *n*-queens problem.

a) Consider *incremental formulation*, try to find a solution using:

- i. *backtracking*,
- ii. *backtracking with forward checking*, and
- iii. *backtracking with forward checking* with the use of the *MRV* and *least-constraining-value (LCV)* heuristics.

Construct a table of average run times for each algorithm for values of *n* up to the largest you can manage.

Comment on your results.

b) Consider a formulation like the one discussed in Section 4.1, Figure 4.3.

- i. Propose a state representation
- ii. Propose an *initial state* that you think would make the run time faster. Justify.
- iii. Propose the *actions (tweaking)* that you think would make the run time faster. Justify.
- iv. Propose heuristics for making the moves. You may use the *min-conflicts (number of attacks)*.
- v. Solve using Simulated Annealing.

Add to the table in (a) the average run times for values of *n* up to the largest you can manage. Comment on your results.

Bonus (10pts):

c) Similar formulation to the one in (b), solve using Genetic Algorithms.

Add to the table in (a) the average run times for values of *n* up to the largest you can manage. Comment on your results.

Notes:

- This is an *individual-type* assignment, no group work is accepted.
- Submit the Report and the Code (as Appendix) in PDF.
- Source code also be submitted in a separate file(s).
- No Zip Files.